Market Cipher B & Compass Momentum

A practical research handbook for Claude

Challenger | Version 1.0 | September 7, 2026

The objective is to find strong companies during constructive pullbacks, then use momentum and flow to improve entry timing. The user wants to capture meaningful portions of emerging trends in individual stocks, with options considered after the underlying stock process is validated.

This handbook summarizes six official Market Cipher lessons, translates the supplied Compass script, and defines how Challenger should investigate the method. It is a private research handoff. It does not establish a profitable strategy or describe verified Watchtower implementation details.

The decision sequence

1. Establish which stocks were eligible and attractive using information available at the decision date.
2. Check market conditions, industry leadership and the stock's own price structure.
3. Identify a pullback with a clear level that would invalidate the trade.
4. Read the momentum-wave sequence and direction of the flow oscillator.
5. Require the chosen confirmation on a completed candle. Record which green dot it is.
6. Define entry, position size and exit before recording a simulated trade.

A green dot is one observation inside this process. It is not a complete stock-selection or trading rule. This is the user's central requirement.

How to read this document

Taught method means a paraphrase of official lessons. **Compass definition** means a formula from the user-supplied script. **Challenger proposal** means our own testable research choice. **Observed result** means an actual local calculation or chart comparison. These categories must remain distinct when Claude learns from the material.

The 16D application is a requested experiment. The sources reviewed do not establish that 16D is optimal for equities. Its candle timing, event frequency and opportunity cost need testing.

Broader V7 coverage has not yet proved better returns. Earlier completed models failed the promotion gates.

Operating boundary

Keep Challenger independent. Do not call, change, deploy, restart or reconfigure Watchtower, Watchtower MCP or their Supabase project. This handbook grants no broker authority. It contains no API keys and requires none for reading.

1. Read the actual Compass display

The following definitions come from the supplied private Pine script [P1]. Its labels alone can be misleading when implementing it in another system.

Display element	What the supplied Compass computes	Interpretation discipline
Light blue wave	A smoothed, normalized price deviation	Read direction, depth and turns in context.
Blue wave	Three-candle arithmetic average of the light wave	Compare successive momentum cycles.
Green dot on the wave	Light wave crosses above blue wave	Any bullish crossover; no oversold requirement.
Bottom Buy dot	The same bullish crossover while light wave is strictly below -60	A narrower event. It is not interchangeable with the wave dot.
Mny Flow	A smoothed price-normalization oscillator	It uses no volume. Do not call it measured institutional buying or actual dollars entering the stock.
VWAP	Light wave minus blue wave	This is not the conventional price-and-volume VWAP used on a price chart.
RSI / Sto RSI	Two smoothed price stochastic calculations, with 40- and 81-candle ranges	Do not replace them with standard RSI and Stoch RSI libraries.

Direction and sign answer different questions

For the user's requested flow-turn filter, **-25, -18, -10** is improving while still negative. **+25, +18, +10** is positive but deteriorating. These are invented teaching numbers, not observed trades. Record both the reading and its change.

A red area becoming less negative may support a recovery hypothesis. A positive reading by itself does not satisfy a requirement that flow be turning upward. Do not infer actual capital flows from either example.

The unresolved dot definition

The user has not yet identified which of the two dot types the named Watchtower 16D methodology uses. Store both flags. Any study that assumes the bottom Buy dot must state that assumption. No Watchtower code or private implementation was inspected to resolve this.

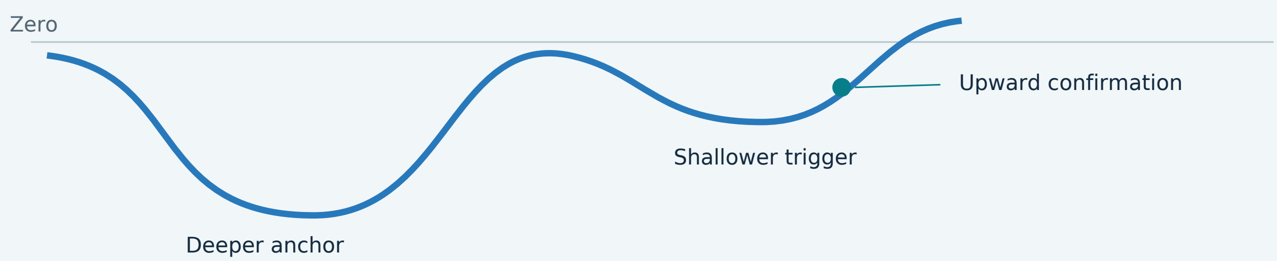
2. The taught setup: waves, flow and context

Core recovery pattern [S1]

The basic lesson describes a substantial negative **anchor wave**, followed by a **shallower trigger wave**. The trigger's upward turn matters; a deep oversold dot can mark the earlier anchor rather than the eventual entry. The flow curve should recover coherently, with a rounded upward shape. It can still be red. Choppy flow or little distinction between the waves weakens the illustrated setup. Ambiguous patterns can require another wave or no trade. [S1, 0:41-4:28; 6:44-7:29; 11:31-12:50]

The words "substantial," "shallower" and "rounded" are visual judgments in the lesson. Numeric definitions in a backtest are additional research choices, not formulas supplied by the teacher.

SCHEMATIC: a recovery sequence, not a price forecast



Multiple timeframes [S2]

The confirmation lesson shows lower-timeframe entries that conflict with falling higher-timeframe momentum and flow. A favorable outcome despite that conflict does not validate the entry process. It recommends waiting for broader support and using the smaller chart to time execution. It also discusses reducing exposure when the entry timeframe weakens while the larger bias remains constructive. [S2, 4:55-9:08; 11:21-13:01]

For Challenger, using 16D as context and daily candles for timing is an adaptation to test. The lesson's example intervals do not prove that this pairing is best.

Pullbacks within an uptrend [S5]

The uptrend lesson begins by establishing the larger trend, then looks for lower-timeframe oversold momentum, easing negative flow and nearby price support. It uses separate support/resistance tools to locate invalidation. Its price-chart VWAP is not Compass's wave-difference plot. It treats weakening momentum near higher prices as a reason to review profits, while noting that a minor warning need not end an otherwise intact trend. [S5, 0:08-3:18; 4:14-6:42; 8:57-10:27]

These are crypto trading examples. Their leverage, position sizes and advertised profits are not copied into Challenger.

3. Divergence, extremes and exits

Distinguish the patterns [S4]

Compare price and oscillator at aligned observation times, using a consistent definition of highs or lows. Do not choose one price swing and an unrelated oscillator swing merely because the lines look favorable.

Pattern	Price	Oscillator	Hypothesis to investigate
Regular bullish divergence	Lower low	Higher low	Selling momentum may be weakening.
Hidden bullish divergence	Higher low	Lower low	A pullback may resolve with the uptrend.
Regular bearish divergence	Higher high	Lower high	Upside momentum may be weakening.
Hidden bearish divergence	Lower high	Higher high	A bounce may resolve with the downtrend.

An anchor followed by a shallower wave is not automatically divergence: price must be compared too. Divergence does not establish the size or timing of a subsequent move. [S4, 0:23-4:12; 5:52-11:35; 15:01-16:33]

Implementation choice: if a pivot needs two later candles to confirm, the signal becomes available only after those candles. Backdating it to the pivot would use future information. The current Compass diagnostic does not implement a price-divergence detector.

Extreme readings are context [S3]

The extremes lesson discusses the +60 and -60 wave regions. Strong trends can remain extreme, and dots can appear before continuation. A reading alone cannot tell how much farther price will move. Smaller charts can cool while the larger chart remains strong. Do not use "overbought" as an automatic short instruction or "oversold" as an automatic long instruction. [S3, 0:43-1:53; 4:05-6:37]

What the exit lesson actually teaches [S6]

The basic exit lesson frames its recovery trade as entering during red flow and taking profit during green flow. In its example, it takes a partial exit and later closes the remainder as price becomes sharp and unstable. It accepts missing the exact top. The trade used a 30-minute chart and lasted several days. [S6, 1:33-4:40; 6:10-7:26]

This does not supply one mechanical exit for every asset or holding period. A first positive flow reading, a deterioration signal and a price-structure exit are different hypotheses. Challenger must register which one is being tested; it must not switch among them after seeing the outcome.

4. Select the stock before selecting the entry

Challenger proposal. These are equity research rules, not a claim that Market Cipher teaches this exact stock-ranking system.

Start with a dated, investable universe

Use the common stocks actually listed at the time, including later failures, mergers and delistings. The existing broad-universe work uses a $10 minimum price and $10 million mean daily dollar volume over 20 sessions, with sufficient valid observations. Keep corporate identities and stock splits consistent. A list of today's winners cannot reconstruct yesterday's opportunity set.

Rank leadership and inspect the pullback

Compare medium-term performance with SPY and with industry peers where dated classifications exist. Look for a stock that has demonstrated leadership, then pulled back without destroying the price structure supporting the trade. Record the prior advance, pullback depth and duration, support level, distance to that level and nearby overhead resistance.

The main V7 model currently studies a rising 50-day average, positive 63-session relative return and a 55-session closing breakout. Its breakout rule is not automatically suitable for a pullback entry. A future V8 adapter should reuse dated universe and leadership information without requiring a fresh breakout at the pullback low. That adapter is not yet implemented.

Use fundamentals with their release dates

Potential diagnostics include revenue growth, margin direction, cash generation, dilution and balance-sheet pressure. Use filings and earnings information available then. Missing data must remain missing; it is not evidence of poor quality. Keep financial diagnostics separate from the baseline until their incremental value is measured. Current broad fundamental and historical industry coverage is incomplete.

A repeatable decision record

Decision	Minimum evidence to record
Eligible	Historical listing, identity, price and liquidity checks.
Attractive	Relative leadership, industry context if available, dated fundamentals.
Constructive pullback	Price level that supports the thesis and what would invalidate it.
Indicator confirmation	Actual values, wave dates, dot type and candle-close availability.
Tradable	Entry rule, position size, costs, exit rule and sufficient room before resistance.
Reject or wait	Explicit failed conditions or missing evidence.

There is no requirement to fill every portfolio slot. A rejected stock remains in the research log so later analysis can compare selected and rejected opportunities.

5. Test on 16D with confirmed data

TradingView expresses "16D" as a day-based chart interval [S7]. That notation alone does not establish the session boundaries required to reproduce its bars elsewhere.

Required data alignment

Match symbol identity, exchange calendar, chart type, adjustments, session settings and candle opening/closing times. Standard candles are the initial comparison target. Do not invent groups of 16 observations or an arbitrary calendar anchor and assume they match TradingView. The September 7 export verifies displayed opening labels for TSLA, but a complete Polygon aggregation calendar is still pending.

Only a completed 16D candle can supply a confirmed historical setup. Its final values must not be applied to earlier daily candles inside that same period. TradingView documents the distinction between developing higher-timeframe values and confirmed values [S8]. For a daily entry, use the latest 16D state that was already complete at that day's decision time.

History needed by this exact script

Component	Earliest possible valid reading under non-degenerate input
Light wave	Candle 2 with the supplied EMA initialization.
Blue wave	Candle 4.
RSI-labelled price stochastic	Candle 41.
Money flow	Candle 68.
Sto RSI-labelled price stochastic	Candle 82.

These counts are chart candles, not daily sessions. On 16D, full oscillator history requires years. Missing or flat input can extend the requirement. EMA agreement can require additional burn-in, as the TSLA comparison demonstrates.

Consequently, the recent broad daily download cannot by itself support a mature 16D flow history. Young companies may be ineligible for this experiment while remaining eligible for the main trend model. Do not make 16D history a universal veto against newly emerging businesses.

Separate experiments to register

A. Selective bottom-dot study: selected stocks, a qualifying pullback, a completed 16D bottom Buy dot and improving flow.

B. Wave-sequence study: selected stocks, a 16D anchor/trigger recovery and improving flow, followed by a completed daily entry condition. The shallower trigger need not print a bottom Buy dot.

Freeze daily entry, setup expiry, selection and exit rules before measuring returns. Raw-dot counts are diagnostic only; buying every dot is excluded.

6. Make the test reproducible

Challenger proposal, version 0.1.0. The current implementation is a wave/flow diagnostic. It neither selects stocks nor places simulated or live orders.

Current diagnostic definition

1. A bearish wave segment starts at an observed downward wave crossover and ends at a subsequent upward crossover.
2. Compare the two most recently completed negative segments when the second completes. The first blue-wave low must be below -60.
3. The second blue-wave low must remain below zero and have no more than 80% of the first low's absolute depth.
4. Over the last four flow readings, the last two changes must be positive. Net rise divided by the sum of absolute changes must be at least 0.75.
5. Separately record whether the fast price stochastic exceeds the slow one and is rising. It is a diagnostic, not a mandatory gate in this version.

The 80%, four-reading and 0.75 choices make a visual idea measurable. They are not vendor parameters or proven optima. This version does not quantify anchor breadth, detect price divergence, verify stock support or implement the daily entry. Do not represent a passing diagnostic as a fully qualified trade.

Entry, sizing and exit specification still required

Before replaying a candidate strategy, register a precise entry trigger and the price-based invalidation rule. Decide how gaps are handled, when stale setups expire, how concurrent signals are ranked and how many positions may be held. Any stop must use executable prices; a gap can cause a larger loss than its planned distance.

A proposed share count is the smaller of (cash allocation divided by entry price) and (risk budget divided by the positive entry-to-invalidation distance), rounded down. This still requires a chosen risk budget, concentration limits and execution assumptions. Do not silently borrow money or increase size after a losing move.

Comparisons that can reveal an incremental edge

Compare the same selected-stock process with and without the Compass filter, selective bottom-dot entries and anchor/trigger entries. Evaluate net compounded account returns, drawdown, volatility, time invested, turnover and trade counts against dividend-reinvested SPY and a feasible matched-stock control. Report base and stressed trading costs.

Previously inspected periods are development data. Keep unsuccessful configurations and all tried variants. Use new validation periods and paper trading after the rules are frozen. Options require a separate executable-quote study; directional accuracy in shares does not establish option profitability. The historical options quote endpoint was denied under current access, so a quote-verified options claim is unavailable.

7. Formula reference for Claude

Compass definition [P1]. These formulas describe the supplied reconstruction. No protected Market Cipher source was obtained or reproduced. Preserve the source fingerprint when implementing a port.

Momentum waves

Let P = (high + low + close) / 3. The custom EMA initializes at its first available input. At each later available input x, it updates state by alpha times (x - state), with alpha = 2 / (length + 1). Missing input emits a missing result while retaining the prior state.

- Center = custom EMA(P, 9).
- Deviation = custom EMA(abs(P - Center), 9).
- Normalized = (P - Center) / (0.015 x Deviation), unless Deviation is zero.
- Light wave = custom EMA(Normalized, 12).
- Blue wave = arithmetic mean of the current and two preceding light-wave values, requiring all three.
- Wave difference, labelled VWAP = Light wave - Blue wave.

Crossovers and dots

An upward crossing requires current Light > Blue and previous Light <= previous Blue, with all four values available. A downward crossing reverses those comparisons. The bottom Buy requires the upward crossing and current Light < -60. Equality at -60 does not satisfy the bottom Buy rule.

Flow calculation

Flow Center = SMA(P, 5). Flow Deviation = SMA(abs(P - Flow Center), 5). Normalize (P - Flow Center) by 0.015 x Flow Deviation, leaving the result missing when the denominator is zero. Money Flow = SMA of those normalized values over 60 available observations. The SMA follows Pine's handling of missing observations. Volume is absent from the calculation.

Price stochastic pair

For each length L in {40, 81}, Raw = 100 x (close - lowest low over L candles) / (highest high over L candles - lowest low over L candles). Require L bars and a positive range. The displayed output is the mean of current Raw and previous Raw, requiring both. The 40 output is labelled RSI; the 81 output is labelled Sto RSI.

Verification discipline

Do not substitute generic library indicators based on the plot names. Verify numerical values, missing-value behavior, equality conditions and dot timing against exported chart observations. A matching recent chart does not validate every initialization regime or symbol.

Private source SHA-256:

`cff8b14fb5f59b31749cef10ada475d30153b5018326dcabe5c6fc2495ccea14`

8. What has actually been verified

Observed result, September 7, 2026. We exported 260 candles from the separate TSLA indicator-testing chart in TradingView. Both Market Cipher B and Compass M were present. The chart used BATS:TSLA, 16D standard candles and enabled dividend adjustment. We restored the testing chart to its previous 5-minute interval and three-chart layout afterward.

Numerical comparison

The local port matched all six supplied Compass numeric series wherever available, including missing-value locations, within an absolute tolerance of 0.00000001. The largest numeric difference was approximately 0.000000000000108. This is formula calibration, not a trading-performance result.

Series	Shared original/Compass observations	Largest original/Compass difference
Light wave	233	11.0087
Blue wave	231	9.4514
Wave difference	231	1.5033
Money flow	193	0
RSI-labelled series	220	Below 0.000000000000015
Sto RSI-labelled series	179	Below 0.000000000000015

Original wave output begins later than Compass output in this sample. The early differences decay; using the strict comparison tolerance above, the last light-wave mismatch has a November 20, 2019 opening label and the last blue-wave mismatch December 13, 2019. This does not mean all earlier discrepancies were economically material. It does mean universal exact equivalence is unproven.

What this sample cannot prove

Neither indicator emitted a positive bottom Buy event in this export. Matching zero events does not validate positive Buy-event parity. The opening labels span June 15, 2010 to August 24, 2026; these are bar labels, not confirmed trade timestamps. The final candle may still be developing and was used only for formula comparison.

This is one symbol and one chart configuration. Complete candle-boundary reconciliation to Polygon, more-symbol validation and a portfolio backtest remain pending. No candidate stock list or return estimate is justified by this calibration alone.

9. Claude handoff and learning protocol

Instructions for the receiving assistant

Use this versioned reference for selective entries in strong stocks. Distinguish official teaching, supplied formulas, proposed rules and measured outcomes. Missing data is unknown. Never invent levels, filing dates, fills or results.

Read the private Compass source if separately provided. Do not assume access to referenced local files. Keep code, data and memory work in Challenger; do not access Watchtower or Watchtower MCP to fill gaps.

When evaluating a candidate, report:

1. **Identity and time:** issuer, symbol, feed, decision date and completed candles used.
2. **Selection:** dated eligibility, liquidity, leadership and fundamental availability.
3. **Price:** advance, pullback, support, resistance and invalidation.
4. **Indicators:** wave dates/values, flow changes, dot type and aligned divergence if present.
5. **Decision:** reject, wait or qualify under a named rule version; identify missing conditions.
6. **Trade:** entry, size, costs, exit and expiry, if those rules are defined.
7. **Outcome:** net return and benchmark comparison; preserve the original decision.

Learn from the process without rewriting history

Keep an append-only journal for selected and rejected stocks, with input snapshots and source hashes. Add later outcomes separately. Investigate selection, timing, invalidation, execution and missing data; a losing trade alone does not identify its cause.

Review a defined batch of outcomes before proposing revisions. Version every change. Use development data for diagnosis and uninspected observations for validation. Do not move thresholds silently until a historical winner appears.

Check comprehension before implementation

- Does a positive but falling flow reading meet an upward-turn rule? **No.**
- Are both green-dot types the same event? **No.**
- Is a shallower momentum low automatically bullish price divergence? **No.**
- May the final 16D value be used on earlier daily candles inside it? **No.**
- Does a matching indicator calculation establish profitable options trades? **No.**
- Must a stock with insufficient 16D warm-up be excluded from every Challenger strategy? **No.**

Future sessions should read this handbook through the project-memory reference. The PDF can be given directly to Claude; it does not retrain a model or transfer itself automatically.

10. Sources and audit trail

Six official lesson transcripts were read in full. English captions were automatically generated and may contain transcription errors. The handbook paraphrases their teaching without endorsing advertised performance.

S0. Official tutorial directory

Market Cipher tutorial videos. Used to establish the lesson links and publisher.

S1. Basic / Core Strategy

Official lesson. Anchor and trigger: 0:41-1:02; flow and entry: 1:45-4:28; waiting: 6:44-7:29; weak examples: 11:31-12:50.

S2. Core Strategy Correctly

Official lesson. Timeframe conflict and confirmation: 4:55-9:08; position management and patience: 11:21-13:01.

S3. Extreme Conditions

Official lesson. Extreme readings: 0:43-1:53; uncertainty and timeframe context: 4:05-6:37.

S4. Divergent Price Action

Official lesson. Aligned comparisons: 0:23-2:04; hidden bullish: 2:21-4:12; regular bullish: 5:52-7:17; pattern distinctions: 7:38-11:35; limits: 15:01-16:33.

S5. Entering Longs in an Uptrend

Official lesson. Trend and pullback: 0:08-3:18; support and invalidation: 4:14-6:42; profit warnings: 8:57-10:27.

S6. Basic Strategy, Part 3: How to Exit

Official lesson. Flow and partial exits: 1:33-4:40; accepting imperfect exits and timeframe: 6:10-7:26.

S7-S8. TradingView documentation

Timeframes and Repainting. Interval notation and confirmed higher-timeframe information.

P1. User-supplied private Compass Momentum Pine script

Received in the September 7 research conversation; fingerprint in section 7. The source itself describes an independent reconstruction with unverified cases. It is stored privately, outside version control.

Local evidence: reports/compass-v8/calibration.json; config/compass-v8.json; tests/test_compass_v8.py. The raw TSLA export is private. Its SHA-256 is:

`240134a48a5883705ae791db89673017e3db59bd2dab38c795fc35610194faf8`

Research cutoff: September 7, 2026. Revise as validation progresses.